Vansh Kulshrestha

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Education

Vellore Institute of Technology – AP Amaravati, Andhra Pradesh
B.Tech in Computer Science and Engineering (Specialization: AI & ML) *Sep 2023 – May 2027*

Skills

AI/ML: Machine Learning, Deep Learning, NLP, Transformers, Spiking Neural Networks, Model Evaluation, Feature Engineering, Data Preprocessing
LLMs & GenAI: LLM Pre-training, RAG, AI Agents & Tool Use, LLM-as-a-Judge Evaluation, Prompt Engineering, Adversarial ML & AI Security
Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NumPy, Pandas, SpikingJelly, FastAPI, Streamlit, Seaborn, Mongoose
Languages: Python, Java (Intermediate), JavaScript (Basic), SQL (Basic)
Databases: Oracle AI Database, MongoDB
Tools & Deployment: Git, GitHub, Docker, Render, Vercel

Experience

Machine Learning Intern – Fox Trading Bangalore, India
Project: Parking Availability Predictor *Mar 2024 – Apr 2024*

- Built a machine learning model to predict parking availability from historical occupancy data, working alongside senior analysts to refine the approach.
- Engineered time-based features (hour, day of week, recurring demand patterns) and compared multiple regression models to select the best performer.
- Validated the model on held-out time periods to confirm accuracy and reliability on unseen data.

Artificial Intelligence Intern – Personifwy Bangalore, India
Project: 500M-Parameter Language Model Training *Mar 2024 – Apr 2024*

- Trained a 500M-parameter transformer model end to end, covering data preparation, tokenization and the training pipeline.
- Ran GPU training with checkpointing, monitoring loss curves to tune learning rate and batch size for stable convergence.
- Evaluated checkpoints on held-out data to select the final model for the team's customer-personalization work.

Research

Detecting Model-Extraction Probing with Semantic Embeddings and MMD *Manuscript in preparation*

- Designing an inference-API wrapper that detects model-extraction and inversion-style probing in LLM traffic using maximum mean discrepancy (MMD) over semantic query embeddings, calibrated on benign traffic only.
- Formulated the threat model and a session-level risk profile with a five-level graduated response to minimize false positives on legitimate users.
- Benchmarking against PRADA, SEAT and other baselines on a 14-pair extraction benchmark.

Projects

Mini-infer – LLM Inference Engine from Scratch – Website *Python, PyTorch, FastAPI*

- Built a Qwen2.5 serving engine with a paged KV cache, continuous batching and a streaming OpenAI-style API, matching HF token for token.
- Reached 2.4× Hugging Face's best throughput (388.9 vs 165.0 tokens/s, 64 concurrent users, ShareGPT traffic, Tesla T4); first-token latency fell from 39 s to 0.18 s at 32 users.
- Cut wasted KV-cache memory from 80.7% to 1.9% with 16-token paged blocks; also implemented INT8 weights (36% less memory) and speculative decoding.

GRPO Reasoning Trainer from Scratch – Website *Python, PyTorch, LoRA, vLLM*

- Implemented GRPO without RL libraries (LoRA, vLLM rollouts), raising Qwen2.5-0.5B's GSM8K test accuracy from 49.1% to 53.3% ($p = 0.001$); adapter released on Hugging Face.
- Ran 6 ablations (group size, KL, format reward, advantage normalization) and screened 1,297 answers for reward hacking.
- Traced a +5-point validation gain on Qwen2.5-1.5B that missed the test set to likely contamination; caught a hidden collapse.

ResearchPilot – Autonomous Research Agent – Live demo *Python, FastAPI, Streamlit, Tavily, Docker*

- Built a plan-and-execute research agent with web search, page retrieval and calculator tools; on the first 22 of 45 benchmark runs it scored 95.5% (automatic grader) and 90.9% (LLM judge), with 97.8% of tool calls succeeding.
- Added prompt-injection defenses, tool error handling and loop limits; improved prompts through LLM-judge failure analysis.

Spiking Neural Network from Scratch *Python, PyTorch, NumPy, SpikingJelly*

- Implemented Leaky Integrate-and-Fire (LIF) neurons with rate, temporal and population spike encoding in NumPy.
- Scaled to a GPU-accelerated multi-layer SNN: ∼95% accuracy on MNIST, 90–95% on N-MNIST and 60–75% on CIFAR10-DVS.
- Analyzed trade-offs between accuracy, temporal representation and compute cost across static and event-based datasets.

Certifications

- Oracle Autonomous AI Database Certified Professional – Certificate